

F-invariants (F1–F6)

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The *F-invariants* (F1–F6) are six discrete behavioural probes the project runs to check whether claims it makes about its own functioning actually hold. Each probe is a yes/no question the project can answer mechanically: F1 — does the agent attribute commits correctly? F2 — can it open pull requests? F3 — are its credentials consistent across tools? Failures don't punish; they narrow what the project claims about itself until the probe passes again — see falsifier-with-grace.

Where named

The F-invariant series is defined across the integrity-check spec and several memos. F1–F4 are operational; F5 and F6 are voice-shape probes.

Related

- Falsifier-with-grace
- Integrity check
- Portability probe

Detail

- **F1 — commit attribution.** Every commit is attributed correctly (author + co-authored-by lines as configured).
- **F2 — PR posting rights.** The agent can open and update PRs via the configured [gh](#) token.
- **F3 — credentials consistency.** Tokens and keys resolve consistently across MCP, [gh](#), and direct API calls.
- **F4 — ledger integrity.** The memo index and substrate ledgers are internally consistent (no orphan IDs, no missing files).
- **F5 — voice density floor.** The agent's writing maintains a minimum density (not collapsing into hedge or boilerplate).
- **F6 — external touch ratio.** Bounded ratio of external (network, third-party) writes to substrate writes; an early signal of drift.

F-invariants are checked daily by the integrity probe and surfaced in `${VADE_CLOUD_STATE_DIR}/integrity-check.json`. A degraded invariant is not punitive; the chain reads the JSON and adapts what it claims about itself.